Clean Hardware Business Plan

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1 Overview

We sell devices with modified firmware, especially ThinkPad laptops. Interested parties include programmers, Free Software advocates, users and organisations concerned about backdoors or spying in device firmware.

Librebooted laptops have improved security capabilities, thanks to installing a Free bootloader in the BIOS chip, and can boot encrypted /boot partitions. They are guaranteed to function fully with 100% free drivers in the FSF certified Parabola or Trisquel linux distributions, making them the best Linux laptops available.

Librebooting devices is a much more involved skill than many users are willing to develop, since the most powerful supported devices require complete disassembly, a BIOS update, some shell commands, and CPU repasting. However, Trisquel is a very easy to use operating system based on Ubuntu, and many individuals want that privacy and security.

2 Pricing

To test the market, some T500 thinkpads were purchased from eBay sellers at about $70 each. Parts were upgraded, raising the price to approximately $200 per laptop. They were sold for $405 plus shipping.

The two competitors have their own internet shops for users to purchase similar devices at $750, and are often out of stock. We plan to

compete with their shops by offering private purchasing with Monero, and a higher volume of lower cost devices sourced from computer recyclers.

We will produce multiple models at two tiers, to maximize profit through price discrimination. Competitors waste valuable materials by only selling top-tier products.

2.1 T500 TOP-TIER PRODUCT $750

The CPUs will be upgraded with a quad-core modification and soldered wire, or a 3.06Ghz dual-core will be used not offered by competitors. 8gb RAM, 256 Gb SSD and 9-cell battery. $750 (match competitor's price, but ours is better). Good device plastics. 1650x1050 or better screen resolution.

2.2 T500 LOW-TIER PRODUCT $500

2.54 to 2.93 Ghz CPUs, 4Gb RAM, 6-cell battery, 120Gb SSD. 1280x800 or better screen resolution.

The replaced CPUs and RAM from top-tier boards can be repurposed for low-tier sales.

3 Shop features

Our shop will offer debit payments via Stripe and Paypal, but also offer Monero and BTC. We expect these crypto payments to see significant use considering the market, and competitors don't offer them.

We will also offer the ability to order computers to be built for a significant markup on each changed option. Competitors do this, but they are wasteful by requiring buyers to ship their own computers to the UK to be worked on. Instead, we will have buyers select from our currently unimproved inventory.

4 Complementary and Supplementary Goods

Replicant OS "phones", which also have free firmware, are often desirable. They don't have cellular capabilities, are slow, and are very inconvenient to use. However, some users will want these.

- Cost: $70
- Sale: $150

Because Librebooted machines add additional cost, many users will opt to purchase proprietary ThinkPads and other devices with reasonably good Linux compatibility. This is why a competitor offers to libreboot machines mailed in for $150 and 2-way shipping to the UK. We can compete by offering the same service in the US (with lower shipping for Americans).

5 Getting The Word Out

Currently, many individuals know about Libreboot and Replicant because they are the only truly "Free Software" machines, and the only machines without three letter agency backdoors. Probably the biggest popularizer is Luke Smith (https://lukesmith.xyz) who has YouTube videos about them.

Because the current pool of customers is highly knowledgeable, it seems that a referral kickback program for people to put on their websites might be a good method.

A lot of free advertising is probably also available through Free Software oriented groups like the FSF who might feature us in their materials.

6 Success Metrics

- More than half of market share in Libreboot laptops sales/month.
- *or* More than 200 sales per month.

7 Obstacles to Success

7.1 Labour costs

The most powerful laptops (T500 or W500) cost 2-5 hours to produce. With some special methods, it may be possible to reliably produce them at 2 hours.

- Firmware update time: 30 minutes
- Disassembly: 30 minutes
- Re-pasting, part upgrades, soldering a quad-core mod wire: 30 minutes
- Reassembly: 30 minutes

7.2 Small-time competitors

We will compete with small producers by being high-volume and developing a trustworthy brand. We'll also offer a very easy buying expericence with our shop and inventory.